

Reasoning Behavior Analysis Report

EXPERIMENTAL SETUP

=====

Data Folder: data/MATH500/deepseek/p1
Total Items: 500
Total Strategies: 9

OVERALL ACCURACY

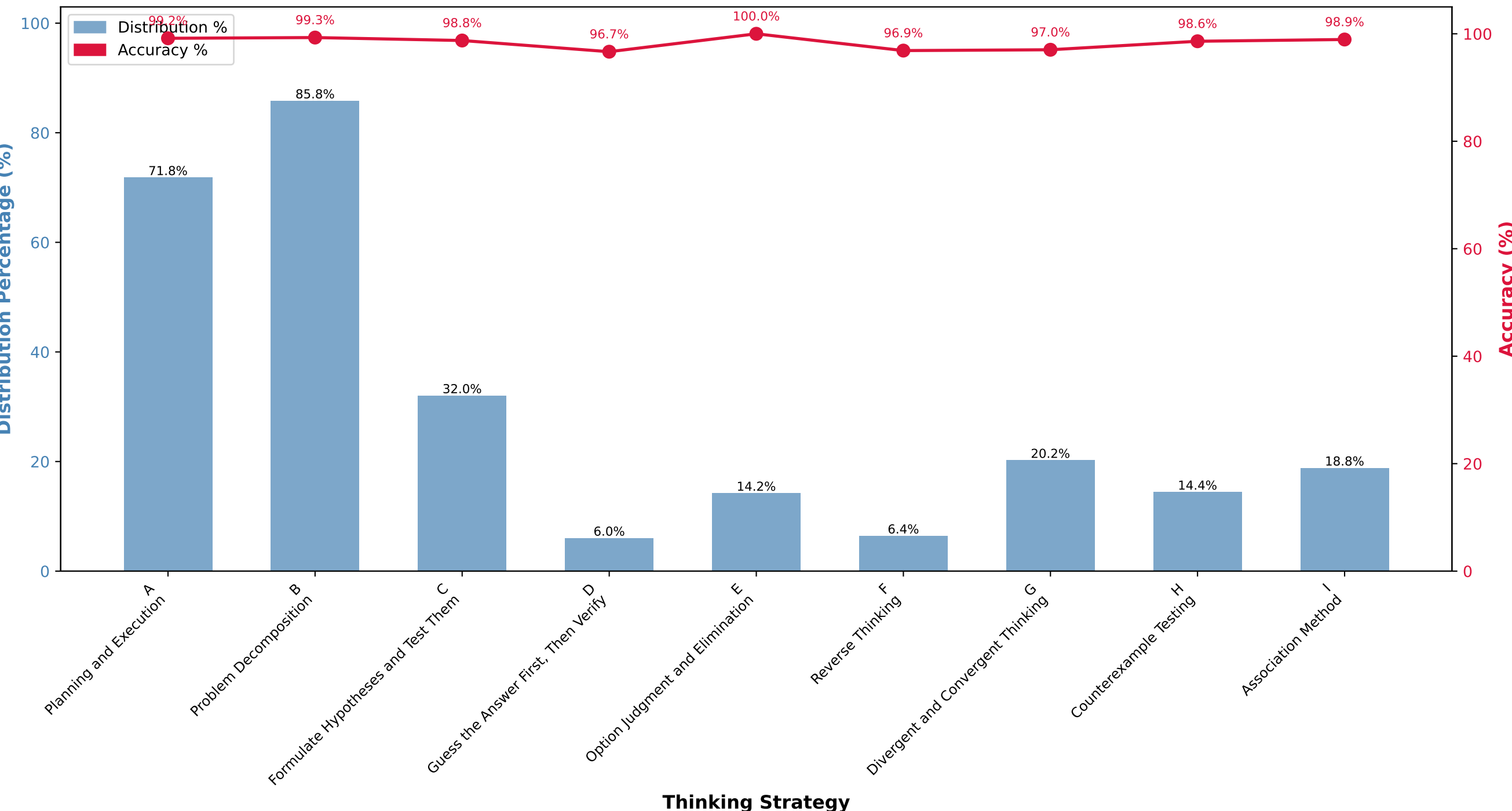
=====

Total Problems: 500
Correct: 496
Incorrect: 4
Overall Accuracy: 99.20%

THINKING STRATEGY DISTRIBUTION

- =====
- A: Planning and Execution
Distribution: 71.8%
Accuracy: 99.16% (356/359)
 - B: Problem Decomposition
Distribution: 85.8%
Accuracy: 99.30% (426/429)
 - C: Formulate Hypotheses and Test Them
Distribution: 32.0%
Accuracy: 98.75% (158/160)
 - D: Guess the Answer First, Then Verify
Distribution: 6.0%
Accuracy: 96.67% (29/30)
 - E: Option Judgment and Elimination
Distribution: 14.2%
Accuracy: 100.00% (71/71)
 - F: Reverse Thinking
Distribution: 6.4%
Accuracy: 96.88% (31/32)
 - G: Divergent and Convergent Thinking
Distribution: 20.2%
Accuracy: 97.03% (98/101)
 - H: Counterexample Testing
Distribution: 14.4%
Accuracy: 98.61% (71/72)
 - I: Association Method
Distribution: 18.8%
Accuracy: 98.94% (93/94)

Thinking Strategy Distribution and Accuracy



Example for Strategy A: Planning and Execution

PROBLEM

If $|x+5|-|3x-6|=0$, find the largest possible value of x . Express your answer as an improper fraction.

GROUND TRUTH ANSWER

$\frac{11}{2}$

MODEL RESPONSE

Step 1: Set Up the Equation

Given:

$$|x + 5| - |3x - 6| = 0$$

Add $|3x - 6|$ to both sides:

$$|x + 5| = |3x - 6|$$

Step 2: Solve the Absolute Value Equation

The equation $|A| = |B|$ implies either:

- $A = B$, or
- $A = -B$

Case 1: $x + 5 = 3x - 6$

Subtract x from both sides:

$$5 = 2x - 6$$

Add 6:

$$11 = 2x$$

Divide by 2:

$$x = \frac{11}{2}$$

Case 2: $x + 5 = -(3x - 6)$

Simplify the right-hand side:

$$x + 5 = -3x + 6$$

Add $3x$ to both sides:

$$4x + 5 = 6$$

Subtract 5:

$$4x = 1$$

Divide by 4:

$$x = \frac{1}{4}$$

Step 3: Determine the Largest Value

The two solutions are:

- $x = \frac{11}{2}$
- $x = \frac{1}{4}$

Since $\frac{11}{2} > \frac{1}{4}$, the largest possible value ...

CHAIN OF THOUGHT (First 1500 chars)

We are given: "If $|x+5|-|3x-6|=0$, find the largest possible value of x . Express your answer as an improper fraction." So equation: $|x+5| - |3x-6| = 0$, so $|x+5| = |3x-6|$. Then we want the largest possible value of x .

Solve: $|x+5| = |3x-6|$. Then square both sides? Alternatively, consider cases. But careful: The equation $|A| = |B|$ implies $A = B$ or $A = -B$.

Case 1: $x+5 = 3x-6 \rightarrow 5+6 = 3x - x \rightarrow 11 = 2x \rightarrow x = 11/2 = 5.5$.

Case 2: $x+5 = -(3x-6) \rightarrow x+5 = -3x+6 \rightarrow 4x = 1 \rightarrow x = 1/4 = 0.25$.

But then check: Are these valid? They are valid if no restrictions? Actually, the equation is absolute, so these are all solutions. But then "largest possible value" is $11/2$. But wait: There might be also conditions from the absolute definitions? Actually, no: The equation is valid for all x . But then check: For $x = 11/2$, then $x+5 = 21/2$, and $3x-6 = 33/2 - 6 = 33/2 - 12/2 = 21/2$, so it's fine. For $x = 1/4$, then $x+5 = 21/4$, and $3x-6 = 3/4 - 6 = 3/4 - 24/4 = -21/4$, so absolute values are $21/4$, so it's fine.

But then "largest possible value" is $11/2$. But then answer as improper fraction: $11/2$ is improper fraction. So answer: $\boxed{\frac{11}{2}}$.

But wait: There is also possibility: if both sides are zero? That would be $x = -5$ and $x = 2$? That gives? Actually, if $x = -5$, then left side: $|0| - |3(-5)-6| = 0 - |-15-6| = -21$, so not zero. If $x=2$, then $|7| - |0| = 7$, so no.

So indeed only two solutions: $x = 1/4$ and $x = 11/2$. So largest is $11/2$.

Thus answer: $\boxed{\frac{11}{2}}$.

Example for Strategy B: Problem Decomposition

PROBLEM

Paul and Jesse each choose a number at random from the first six primes. What is the probability that the sum of the numbers they choose is even?

GROUND TRUTH ANSWER

$\frac{13}{18}$

MODEL RESPONSE

The first six prime numbers are 2, 3, 5, 7, 11, and 13. Among these, only 2 is even, and the rest are odd.

The sum of two numbers is even if both numbers are even or both numbers are odd.

- Probability both choose even: Since only one even number, the probability is $((1/6) \times (1/6) = 1/36)$.
- Probability both choose odd: There are five odd numbers, so the probability is $((5/6) \times (5/6) = 25/36)$.

The total probability that the sum is even is $(1/36 + 25/36 = 26/36 = 13/18)$.

This can also be determined by considering all possible pairs. There are $(6 \times 6 = 36)$ possible pairs. The number of pairs where both numbers are even is 1 (both choose 2). The number of pairs where both numbers are odd is $(5 \times 5 = 25)$. Thus, there are $(1 + 25 = 26)$ favorab...

CHAIN OF THOUGHT (First 1500 chars)

I need to find the probability that the sum of the numbers chosen by Paul and Jesse is even. They each choose a number from the first six primes. First, I should list out the first six prime numbers. Primes are numbers greater than 1 that have no divisors other than 1 and themselves. So, the first six primes are: 2, 3, 5, 7, 11, 13.

Now, each of them chooses one number at random from these six, and since they're choosing at random, each number has an equal chance of being chosen. So, there are 6 choices for Paul and 6 choices for Jesse, meaning there are 36 possible pairs of choices in total.

I need the sum of their numbers to be even. The sum of two numbers is even only if both numbers are even or both numbers are odd. Because even + even = even, odd + odd = even, and even + odd = odd.

So, I need to find the probability that both choose even numbers or both choose odd numbers.

Now, looking at the list of primes: 2, 3, 5, 7, 11, 13. Among these, which are even? Only 2 is even. All the others are odd. So, there is only one even prime and five odd primes.

So, the probability that both choose even numbers: since only one even number, probability Paul chooses even is $1/6$, and probability Jesse chooses even is also $1/6$. Since they choose independently, probability both choose even is $(1/6) * (1/6) = 1/36$.

Now, probability both choose odd: probability Paul chooses odd is $5/6$, and Jesse chooses odd is $5/6$, so both odd: $(5/6) * (5/6) = 25/36$.

Therefore, the total probability t...

Example for Strategy C: Formulate Hypotheses and Test Them

PROBLEM

Let $P(x)$ be a quadratic polynomial with real coefficients satisfying $x^2 - 2x + 2 \leq P(x) \leq 2x^2 - 4x + 3$ for all real numbers x , and suppose $P(11) = 181$. Find $P(16)$.

GROUND TRUTH ANSWER

406

MODEL RESPONSE

Step 1. Express $P(x)$ in General Form

Since $P(x)$ is a quadratic polynomial with real coefficients, write:

$$P(x) = ax^2 + bx + c$$

Step 2. Define the Difference Functions

Given:

$$x^2 - 2x + 2 \leq P(x) \leq 2x^2 - 4x + 3 \quad \text{for all } x \in \mathbb{R}$$

Define:

$$\begin{aligned} Q(x) &= P(x) - (x^2 - 2x + 2) \geq 0 \\ R(x) &= (2x^2 - 4x + 3) - P(x) \geq 0 \end{aligned}$$

Substitute $P(x) = ax^2 + bx + c$:

$$\begin{aligned} Q(x) &= (a - 1)x^2 + (b + 2)x + (c - 2) \\ R(x) &= (2 - a)x^2 + (-4 - b)x + (3 - c) \end{aligned}$$

Step 3. Sum of $Q(x)$ and $R(x)$

$$\begin{aligned} Q(x) + R(x) &= [(a - 1) + (2 - a)]x^2 + [(b + 2) + (-4 - b)]x + [(c - 2) + (3 - c)] \\ &= x^2 - 2x + 1 = (x - 1)^2 \end{aligned}$$

So,

$$Q(x) + \dots$$

CHAIN OF THOUGHT (First 1500 chars)

We are given: "Let $P(x)$ be a quadratic polynomial with real coefficients satisfying $x^2 - 2x + 2 \leq P(x) \leq 2x^2 - 4x + 3$ for all real numbers x , and suppose $P(11) = 181$. Find $P(16)$."

Given: $P(x)$ is quadratic. So $P(x) = ax^2 + bx + c$.

Constraints: For all real x ,
 $x^2 - 2x + 2 \leq P(x) \leq 2x^2 - 4x + 3$.

That means:

- $P(x) - (x^2 - 2x + 2) \geq 0$ for all x .
- $(2x^2 - 4x + 3) - P(x) \geq 0$ for all x .

So define $Q(x) = P(x) - (x^2 - 2x + 2) = (a-1)x^2 + (b+2)x + (c-2)$ is nonnegative for all x .

And $R(x) = (2x^2 - 4x + 3) - P(x) = (2-a)x^2 + (-4-b)x + (3-c)$ is nonnegative for all x .

Since these are quadratic polynomials that are nonnegative for all x , their leading coefficients must be positive (or zero? But then they'd be linear? Actually, if quadratic is nonnegative for all x , then it is either a constant positive or a quadratic with positive leading coefficient and discriminant ≤ 0 . But here it's quadratic? Actually, careful: $P(x)$ is quadratic. So a is nonzero. But then $a-1$ and $2-a$ could be zero? Possibly if $a=1$ or $a=2$? But then these become linear? But then they are not necessarily nonnegative for all x . Actually, if $a=1$, then $Q(x)$ becomes $(b+2)x + (c-2)$. For that to be ≥ 0 for all x , it must be constant and that constant ≥ 0 . But then $b+2=0$, and $c-2 \geq 0$. Similarly, if $a=2$, then $R(x)$ becomes $(-4-b)x + (3-c)$ must be constant and ≥ 0 . So these are borderline cases. But we'll check.

Alternatively, we can also consider that the inequalities hold for all ...

Example for Strategy D: Guess the Answer First, Then Verify

PROBLEM

Let $(a_1, a_2, \dots, a_n)$ be a sequence of positive real numbers, such that $\sum_{i=1}^n a_i = 96$, $\sum_{i=1}^n a_i^2 = 144$, $\sum_{i=1}^n a_i^3 = 216$. Find the sum of all possible values of n .

GROUND TRUTH ANSWER

64

MODEL RESPONSE

Step 1. Analyze the Given Conditions

Let $(a_1, a_2, \dots, a_n)$ be positive real numbers such that:

$$\sum_{i=1}^n a_i = 96, \quad \sum_{i=1}^n a_i^2 = 144, \quad \sum_{i=1}^n a_i^3 = 216$$

Step 2. Apply the Cauchy–Schwarz Inequality

Consider the vectors:

$$\mathbf{u} = (\sqrt{a_1^3}, \sqrt{a_2^3}, \dots, \sqrt{a_n^3}), \quad \mathbf{v} = (\sqrt{a_1}, \sqrt{a_2}, \dots, \sqrt{a_n})$$

By the Cauchy–Schwarz inequality:

$$(\mathbf{u} \cdot \mathbf{u})(\mathbf{v} \cdot \mathbf{v}) \geq (\mathbf{u} \cdot \mathbf{v})^2$$

This translates to:

$$\left(\sum a_i^3\right) \left(\sum a_i\right) \geq \left(\sum a_i^2\right)^2$$

Substitute the given values:

$$216 \cdot 96 = 20736, \quad \text{and} \quad 144^2 = 20736$$

CHAIN OF THOUGHT (First 1500 chars)

We are given: Let $(a_1, a_2, \dots, a_n)$ be a sequence of positive real numbers, such that $\sum a_i = 96$, $\sum a_i^2 = 144$, $\sum a_i^3 = 216$. Find the sum of all possible values of n .

Let's denote: $S_1 = \sum a_i = 96$, $S_2 = \sum a_i^2 = 144$, $S_3 = \sum a_i^3 = 216$. We have positive reals.

Observing pattern: 96,144,216. Possibly constant? If all a_i are equal, then $a_i = \text{constant} = c$. Then $n \cdot c = 96$, $n \cdot c^2 = 144$, so $c = (144)/(96)=1.5$, and then $n = 96/1.5 = 64$. And then check: $n \cdot c^3 = 64 \cdot (3.375)=216$. So indeed if all numbers are 1.5, then $n=64$ works.

But there might be other sequences. Possibly some numbers are different. Let's denote $x = a_i$. Then maybe we can use some inequality like Cauchy-Schwarz? Alternatively, consider that (a_i^3) and (a_i^2) and (a_i) might be related. For each term, $a_i^3 - 1.5 a_i^2$? Actually, note: $216/144 = 1.5$, so average of a_i given squared weights? Alternatively, consider: $a_i^3 - 1.5 a_i^2 = a_i^2(a_i - 1.5)$. Sum gives $216 - 1.5 \cdot 144 = 216 - 216 = 0$. So $\sum a_i^2 (a_i - 1.5) = 0$. Since $a_i > 0$, then for each term, $a_i^2 > 0$. So if any a_i is not equal to 1.5, then there must be some a_j such that $a_j - 1.5$ is negative and some positive. So the numbers are either exactly 1.5 or some are less and some are greater than 1.5. But then also consider: $\sum (a_i - 1.5)^2$? That equals $\sum a_i^2 - 3 \sum a_i + n(1.5^2)$? Actually careful: $(a_i - 1.5)^2 = a_i^2 - 3a_i + 2.25$. So $\sum (a_i - 1.5)^2 = 144 - 3 \cdot 96 + 2.25 n = 144 - 288 + 2.25 n = 2.25 n - 144$. That is not necessarily zero. Bu...

Example for Strategy E: Option Judgment and Elimination

PROBLEM

A strictly increasing sequence of positive integers $a_1, a_2, a_3, \dots$ has the property that for every positive integer k , the subsequence $a_{2k-1}, a_{2k}, a_{2k+1}$ is geometric and the subsequence $a_{2k}, a_{2k+1}, a_{2k+2}$ is arithmetic. Suppose that $a_{13} = 2016$. Find a_1 .

GROUND TRUTH ANSWER

504

MODEL RESPONSE

Step 1. Understand the Conditions

The sequence $(a_1, a_2, a_3, \dots)$ is strictly increasing and consists of positive integers. For every positive integer k :

- The triple $(a_{2k-1}, a_{2k}, a_{2k+1})$ forms a geometric progression.
- The triple $(a_{2k}, a_{2k+1}, a_{2k+2})$ forms an arithmetic progression.

Also, $a_{13} = 2016$. Our goal is to find a_1 .

Step 2. Express the Terms in Terms of a_1 and a Ratio

Let:

$$r = \frac{a_2}{a_1}$$

Since (a_1, a_2, a_3) form a geometric progression:

$$a_2 = r a_1, \quad a_3 = r^2 a_1$$

Now, (a_2, a_3, a_4) form an arithmetic progression:

$$2a_3 = a_2 + a_4 \implies a_4 = 2a_3 - a_2 = 2r^2 a_1 - r a_1 = a_1 r (2r - 1)$$

Next, (a_3, a_4, a_5)

CHAIN OF THOUGHT (First 1500 chars)

We are given: "A strictly increasing sequence of positive integers $a_1, a_2, a_3, \dots$ has the property that for every positive integer k , the subsequence $a_{2k-1}, a_{2k}, a_{2k+1}$ is geometric and the subsequence $a_{2k}, a_{2k+1}, a_{2k+2}$ is arithmetic. Suppose that $a_{13} = 2016$. Find a_1 ."

Let's denote terms: $a_1, a_2, a_3, a_4, a_5, a_6, \dots$

Properties: For every positive integer k :

(1) $a_{2k-1}, a_{2k}, a_{2k+1}$ is geometric. So: $a_{2k}^2 = a_{2k-1} \cdot a_{2k+1}$? Actually careful: Geometric means there is a common ratio r such that $a_{2k} = r \cdot a_{2k-1}$ and $a_{2k+1} = r^2 \cdot a_{2k-1}$? But careful: It says "the subsequence is geometric", so it means that $a_{2k}/a_{2k-1} = a_{2k+1}/a_{2k}$? Actually, if three numbers are in geometric progression, then the middle is the geometric mean: $a_{2k}^2 = a_{2k-1} \cdot a_{2k+1}$. But it doesn't necessarily mean that the ratio is constant? Actually, if a, b, c are in geometric progression, then $b/a = c/b$. So indeed, $a_{2k}/a_{2k-1} = a_{2k+1}/a_{2k}$. So let that common ratio be r_k . But careful: It might be different for different k . So we denote: For k , let $r_k = a_{2k}/a_{2k-1} = a_{2k+1}/a_{2k}$. So then $a_{2k} = r_k a_{2k-1}$ and $a_{2k+1} = r_k^2 a_{2k-1}$.

(2) For every positive integer k , the subsequence $a_{2k}, a_{2k+1}, a_{2k+2}$ is arithmetic. So that means: $2a_{2k+1} = a_{2k} + a_{2k+2}$ or equivalently $a_{2k+2} = 2a_{2k+1} - a_{2k}$. Alternatively, the common difference d_k such that $a_{2k+1} = a_{2k} + d_k$ and $a_{2k+2} = a_{2k+1} + d_k$.

Example for Strategy F: Reverse Thinking

PROBLEM

Suppose that f is a function and f^{-1} is the inverse of f . If $f(1)=2$, $f(2) = 6$, and $f(3)=5$, then what is $f^{-1}(f^{-1}(6))$?

GROUND TRUTH ANSWER

1

MODEL RESPONSE

The inverse function f^{-1} reverses the mapping of f . Given that $f(2) = 6$, it follows that $f^{-1}(6) = 2$.

Then, $f^{-1}(f^{-1}(6)) = f^{-1}(2)$. Since $f(1) = 2$, it follows that $f^{-1}(2) = 1$.

Thus, $f^{-1}(f^{-1}(6)) = 1$.

The value $f(3) = 5$ is not needed for this calculation.

$\boxed{1}$

CHAIN OF THOUGHT (First 1500 chars)

I have this problem: Suppose that f is a function and f^{-1} is the inverse of f . If $f(1)=2$, $f(2)=6$, and $f(3)=5$, then what is $f^{-1}(f^{-1}(6))$?

First, I need to understand what the inverse function does. The inverse function, f^{-1} , reverses the action of f . So, if $f(a) = b$, then $f^{-1}(b) = a$.

Given that $f(1)=2$, so $f^{-1}(2) = 1$.

$f(2)=6$, so $f^{-1}(6) = 2$.

$f(3)=5$, so $f^{-1}(5) = 3$.

Now, I need to find $f^{-1}(f^{-1}(6))$. I know that $f^{-1}(6) = 2$, from above. So, $f^{-1}(f^{-1}(6)) = f^{-1}(2)$.

And from above, $f^{-1}(2) = 1$.

So, that should be the answer. But let me double-check.

$f^{-1}(6) = 2$, and then $f^{-1}(2) = 1$. So, $f^{-1}(f^{-1}(6)) = 1$.

The problem gives $f(3)=5$, but I didn't need that for this calculation. Maybe it's just extra information or perhaps I missed something.

I should ensure that the function is invertible. With the given points, since each input has a unique output, and each output is unique, it should be fine. $f(1)=2$, $f(2)=6$, $f(3)=5$, so all outputs are distinct, so inverse exists for these values.

So, I think I'm good.

Therefore, the answer should be 1.

Example for Strategy G: Divergent and Convergent Thinking

PROBLEM

Find the real roots of

$$\frac{(x+1)(x-3)}{5} + \frac{(x+3)(x-5)}{9(x+4)(x-6)} - \frac{2(x+5)(x-7)}{13(x+6)(x-8)} = \frac{92}{585}$$

Enter the real roots, separated by commas.

GROUND TRUTH ANSWER

$$1 \pm \sqrt{19}$$

MODEL RESPONSE

Step 1. Rewrite the Equation in a Convenient Form

We start with:

$$\frac{(x+1)(x-3)}{5} + \frac{(x+3)(x-5)}{9(x+4)(x-6)} - \frac{2(x+5)(x-7)}{13(x+6)(x-8)} = \frac{92}{585}$$

Notice that each numerator and denominator is a quadratic in (x) . For example:

$$\begin{aligned} (x+1)(x-3) &= x^2 - 2x - 3 = (x-1)^2 - 4 \\ (x+2)(x-4) &= x^2 - 2x - 8 = (x-1)^2 - 9 \end{aligned}$$

Let:

$$u = (x-1)^2$$

Then the terms become:

$$\begin{aligned} \text{First term: } & \frac{u-4}{5} \\ \text{Second term: } & \frac{u-16}{9(u-25)} \\ \text{Third term: } & -\frac{2(u-36)}{13(u-49)} \end{aligned}$$

So the equation becomes:

$$\frac{u-4}{5} + \frac{u-16}{9(u-25)} - \frac{2(u-36)}{13(u-49)} = \frac{92}{585}$$

Step 2. Clear Denominators

Multiply both sides by:

$$585(u-25)(u-49)$$

CHAIN OF THOUGHT (First 1500 chars)

We are given:

Find the real roots of

$$\frac{(x+1)(x-3)}{5} + \frac{(x+3)(x-5)}{9(x+4)(x-6)} - \frac{2(x+5)(x-7)}{13(x+6)(x-8)} = \frac{92}{585}$$

We want: Enter the real roots, separated by commas.

Let's denote equation:

$$(1) \left(\frac{(x+1)(x-3)}{5} \right) + \left(\frac{(x+3)(x-5)}{9(x+4)(x-6)} \right) - \left(\frac{2(x+5)(x-7)}{13(x+6)(x-8)} \right) = \frac{92}{585}$$

Observing pattern: Terms are of form: $\frac{(x+(2k-1))(x-(2k+1))}{(2k+1)(x+2k)(x-2k-2)}$? Actually, check:

First term: $k=1$: numerator: $(x+1)(x-3) = (x+(2*1-1))(x-(2*1+1))$? Actually: For first term: denominator factor: $5 = (2*1+1)$? Actually: $2*1+1 = 3$, but it's 5. So not exactly.

Maybe pattern:

Term1: coefficient: $1/5$, pattern: $(x+1)(x-3)$ and denominator: $(x+2)(x-4)$

Term2: coefficient: $1/9$, pattern: $(x+3)(x-5)$ and denominator: $(x+4)(x-6)$

Term3: coefficient: $-2/13$, pattern: $(x+5)(x-7)$ and denominator: $(x+6)(x-8)$

Maybe there is transformation: Let $u = x-1$? Alternatively, maybe we can simplify each fraction by performing partial fractions? Alternatively, maybe we can combine them into a single fraction? That would yield a high degree polynomial. Possibly there is cancellation.

Alternatively, try to simplify each term:

Consider term: $\frac{(x+a)(x-a-2)}{(2a+1)}$? Actually, pattern: For first term: $a=1$ gives: $(x+1)(x-3)$ and denominator: $(x+2)(x-4)$. Write: $(x+1)(x-3) = (x-1)^2 - 4$? Actually: $(x+1)(x-3) = x^2 - 2x - 3 = (x-1)^2 - 4$.

And denominator: $(x+2)(x-4) = x^2 - 2x - 8 = (x-1)^2 - 9$.

So first term become...

Example for Strategy H: Counterexample Testing

PROBLEM

Let x, y, z be positive real numbers such that $xyz(x + y + z) = 1$. Find the minimum value of $\frac{1}{(x + y)(y + z)}$.

GROUND TRUTH ANSWER

2

MODEL RESPONSE

Step 1. Rewrite the Expression

We wish to minimize $\frac{1}{(x + y)(y + z)}$ subject to the condition $xyz(x + y + z) = 1$, with $x, y, z > 0$.

Expand the expression: $(x + y)(y + z) = xy + xz + y^2 + yz = y(x + y + z) + xz$.
Let $S = x + y + z$.
Then $E = yS + xz$.

Step 2. Use the Given Condition

From the condition: $xyzS = 1 \implies xz = \frac{1}{yS}$.
Substitute into (1): $E = yS + \frac{1}{yS}$.
Let $T = yS$.
Then $E = T + \frac{1}{T}$.

Step 3. Check the Constraints

We must ensure that positive numbers x and z exist such that $x + z = S - y$, $xz = \frac{1}{yS}$.
By the AM–GM inequality: $(x + z)^2 \geq 4xz$.

CHAIN OF THOUGHT (First 1500 chars)

We are given: Let $x, y, z > 0$ such that $xyz(x+y+z) = 1$. Find the minimum value of $\frac{1}{(x+y)(y+z)}$.

We want: minimize $\frac{1}{(x+y)(y+z)}$ subject to condition: $xyz(x+y+z) = 1$.

Let's denote: $\frac{1}{(x+y)(y+z)} = ?$ Expand: $= \frac{1}{xy + xz + y^2 + yz}$? Actually: $\frac{1}{(x+y)(y+z)} = \frac{1}{xy + xz + y^2 + yz}$? Wait careful: $\frac{1}{(x+y)(y+z)} = \frac{1}{x^2y + x^2z + xy^2 + xz^2 + y^3 + yz^2 + y^2z + yz^3}$? That is not correct: $\frac{1}{(x+y)(y+z)} = \frac{1}{x^2y + x^2z + xy^2 + xz^2 + y^3 + yz^2 + y^2z + yz^3}$? Actually: Multiply: $\frac{1}{(x+y)(y+z)} = \frac{1}{x^2y + x^2z + xy^2 + xz^2 + y^3 + yz^2 + y^2z + yz^3}$? That is wrong: Actually: $\frac{1}{(x+y)(y+z)} = \frac{1}{x^2y + x^2z + xy^2 + xz^2 + y^3 + yz^2 + y^2z + yz^3}$? Let's do: $\frac{1}{(x+y)(y+z)} = \frac{1}{x^2y + x^2z + xy^2 + xz^2 + y^3 + yz^2 + y^2z + yz^3}$? That is not: Actually: $\frac{1}{(x+y)(y+z)} = \frac{1}{x^2y + x^2z + xy^2 + xz^2 + y^3 + yz^2 + y^2z + yz^3}$? I'm mixing: Actually: $\frac{1}{(x+y)(y+z)} = \frac{1}{x^2y + x^2z + xy^2 + xz^2 + y^3 + yz^2 + y^2z + yz^3}$? That is if it were $\frac{1}{(x+y)(x+z)}$? No.

Better: $\frac{1}{(x+y)(y+z)} = \frac{1}{x^2y + x^2z + xy^2 + xz^2 + y^3 + yz^2 + y^2z + yz^3}$? Actually: $\frac{1}{(x+y)(y+z)} = \frac{1}{x^2y + x^2z + xy^2 + xz^2 + y^3 + yz^2 + y^2z + yz^3}$? That is not: Actually: $\frac{1}{(x+y)(y+z)} = \frac{1}{x^2y + x^2z + xy^2 + xz^2 + y^3 + yz^2 + y^2z + yz^3}$? Multiply: $\frac{1}{(x+y)(y+z)} = \frac{1}{x^2y + x^2z + xy^2 + xz^2 + y^3 + yz^2 + y^2z + yz^3}$? That is if it were $\frac{1}{(x+y)(x+z)}$ then it's $\frac{1}{x^2 + xz + xy + yz} = \frac{1}{x^2 + x(y+z) + yz}$. But here it's $\frac{1}{(x+y)(y+z)} = \frac{1}{x^2y + x^2z + xy^2 + xz^2 + y^3 + yz^2 + y^2z + yz^3}$? No.

Let's do properly: $\frac{1}{(x+y)(y+z)} = \frac{1}{x^2y + x^2z + xy^2 + xz^2 + y^3 + yz^2 + y^2z + yz^3}$? That is not: Actually: $\frac{1}{(x+y)(y+z)} = \frac{1}{x^2y + x^2z + xy^2 + xz^2 + y^3 + yz^2 + y^2z + yz^3}$? I'm mistaken. Actually: $\frac{1}{(x+y)(y+z)} = \frac{1}{x^2y + x^2z + xy^2 + xz^2 + y^3 + yz^2 + y^2z + yz^3}$? That is if it were $\frac{1}{(x+y)(x+z)}$ then it's $\frac{1}{x^2 + xz + xy + yz} = \frac{1}{x^2 + x(y+z) + yz}$. But here it's $\frac{1}{(x+y)(y+z)} = \frac{1}{x^2y + x^2z + xy^2 + xz^2 + y^3 + yz^2 + y^2z + yz^3}$? Wait: $\frac{1}{(x+y)(y+z)} = \frac{1}{x^2y + x^2z + xy^2 + xz^2 + y^3 + yz^2 + y^2z + yz^3}$? That is not correct. I'll do: $\frac{1}{(x+y)(y+z)} = \frac{1}{x^2y + x^2z + xy^2 + xz^2 + y^3 + yz^2 + y^2z + yz^3}$? Multiply: $\frac{1}{(x+y)(y+z)} = \frac{1}{x^2y + x^2z + xy^2 + xz^2 + y^3 + yz^2 + y^2z + yz^3}$? That is if it were $\frac{1}{(x+y)(x+z)}$ then it's $\frac{1}{x^2 + xz + xy + yz} = \frac{1}{x^2 + x(y+z) + yz}$. But here it's $\frac{1}{(x+y)(y+z)} = \frac{1}{x^2y + x^2z + xy^2 + xz^2 + y^3 + yz^2 + y^2z + yz^3}$? I'll do step by step: $\frac{1}{(x+y)(y+z)} = \frac{1}{x^2y + x^2z + xy^2 + xz^2 + y^3 + yz^2 + y^2z + yz^3}$.

Example for Strategy I: Association Method

PROBLEM

Let z be a complex number such that $z^5 = 1$ and $z \neq 1$. Compute $\left[z + \frac{1}{z} + z^2 + \frac{1}{z^2} \right]$

GROUND TRUTH ANSWER

-1

MODEL RESPONSE

Step 1. Express the Given Expression

We want to compute:

$$\left[z + \frac{1}{z} + z^2 + \frac{1}{z^2} \right]$$

Note that since $(z^5 = 1)$, we have $\left(\frac{1}{z} = z^4 \right)$ and $\left(\frac{1}{z^2} = z^3 \right)$. So the expression becomes:

$$\left[z + z^4 + z^2 + z^3 \right]$$

Step 2. Use the Fact That $(z^5 = 1)$

Since $(z \neq 1)$, (z) is a non-real 5th root of unity. These roots satisfy the polynomial:

$$z^4 + z^3 + z^2 + z + 1 = 0$$

Step 3. Relate the Expression to the Polynomial

Divide the polynomial by (z^2) (valid since $(z \neq 0)$):

$$z^2 + z + 1 + \frac{1}{z} + \frac{1}{z^2} = 0$$

Rewriting:

$$z^2 + \frac{1}{z^2} + z + \frac{1}{z} = -1$$

But this is exactly our expression (E) . Therefore:

$$E = -1$$

...

CHAIN OF THOUGHT (First 1500 chars)

We are given: "Let z be a complex number such that $z^5 = 1$ and $z \neq 1$. Compute $\left[z + \frac{1}{z} + z^2 + \frac{1}{z^2} \right]$ "

Step: z is a primitive 5th root of unity? Actually condition: $z^5=1, z \neq 1$. So z is any of the non-real 5th roots. Then compute: $z + 1/z + z^2 + 1/z^2$. But note that $1/z = \text{conjugate}$? Actually if $z = e^{(2\pi i k/5)}$ then $1/z = e^{(-2\pi i k/5)}$. So then $z + 1/z = 2 \cos(2\pi k/5)$ and $z^2 + 1/z^2 = 2 \cos(4\pi k/5)$. So $\text{sum} = 2(\cos(2\pi k/5) + \cos(4\pi k/5))$. For $k=1$: $\cos(72^\circ) + \cos(144^\circ) = \cos 72 + \cos 144 = (0.309016...) + (-0.809016...) = -0.5$, so $\text{sum} = -1$. For $k=2$: $\cos(144) + \cos(288) = \cos 144 + \cos 72 = -0.809016 + 0.309016 = -0.5$, so $\text{sum} = -1$. For $k=3$: $\cos(216) + \cos(432) = \cos 216 + \cos 72 = -0.809016 + 0.309016 = -0.5$, so $\text{sum} = -1$. For $k=4$: $\cos(288) + \cos(576) = \cos 72 + \cos 144 = 0.309016 + (-0.809016) = -0.5$, so $\text{sum} = -1$. Actually careful: For $k=3$: $z = e^{(6\pi i/5)}$, then $z + 1/z = 2 \cos(6\pi/5) = 2 \cos(216^\circ) = 2 \cos(144^\circ)$ because $\cos(216) = \cos(180+36) = -\cos 36$? Actually better: Actually, it's symmetric: Actually, note that if z is a primitive 5th root, then $z^4 = \text{conjugate of } z$, so $z + 1/z$ is the same for z and z^4 . And similarly z^2 and z^3 . So indeed the sum is always -1. So answer is -1.

But wait: The problem likely expects answer: $\boxed{-1}$.

But let's do algebra: $z^5=1$ so $z^4+z^3+z^2+z+1=0$. Divide by z^2 : $z^2+z+1+1/z+1/z^2=0$, so $z^2+1/z^2 + z+1/z = -1$. So answer is -1.

Thus answer: $\boxed{-1}$.